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Metasurface-Based RF Energy Harvester for 2.6 GHz Wireless Power Transfer Applications.

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Wireless power transfer (WPT) and radio frequency (RF) energy harvesting have emerged as vital technologies for allowing maintenance-free power solutions in modern electronics, particularly within the expanding Internet of Things (IoT) ecosystem. The aim of this research is to develop a metamaterial-based RF energy absorber capable of powering low-energy consumption devices, thereby eliminating the need for conventional batteries and enabling sustainable, continuous operation. This study presents original research on the design, simulation, fabrication, and experimental validation of a metamaterial absorber unit cell targeting ambient RF energy from 4G LTE Band 7 signals operating at 2.6 GHz, aimed at capturing ambient RF energy with high efficiency. A novel metamaterial unit cell consist of T-shaped split-ring resonator, copper reflector and FR-4 substrate layer in the middle was designed and simulated using Ansys HFSS software to achieve strong electromagnetic resonance. Following simulation, the unit cell was fabricated using standard PCB etching techniques, and array configurations of 2×2 and 3×3 were assembled to investigate scalability. Experimental validation was conducted using a spectrum analyser in vector network analyser mode, employing a non-contact setup with a copper ring antenna as the transmitter. Simulated results demonstrated a peak absorptivity of 98.7% at 2.6 GHz, indicating excellent impedance matching and energy capture. Experimental measurements closely matched these findings, with the 3×3 array exhibiting strong absorption characteristics of 94.2% . The study concludes that this metasurface design is effective as an individual unit and can be scaled without significantly declining performance. The proposed metamaterial absorber exhibits notable advantages over conventional RF energy systems, including structural simplicity, cost-effectiveness, and scalability, making it highly suitable for IoT sensors, wearable devices, smart city infrastructure, and remote monitoring applications. Recommendations for future work include incorporating the metasurface absorber with rectifying circuitry to create a complete rectenna, exploring multi-band and broadband configurations, and performing long-term field deployment to assess durability and effectiveness in diverse operational conditions.

Keywords: Internet of Things, Metasurface Absorber, RF Energy Harvesting, Wireless Power Transfer

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